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Spatially blocked pluvial flood learning on the H3 grid: open labels, adaptive refinement, and a rainfall-conditioned cell index

Highlights

- A hierarchical hexagonal grid unifies learning, validation and refinement
- Public flood observations support reproducible pluvial-flood screening
- Class-prevalence baselines change interpretation of classification performance
- Adaptive refinement uses 57% as many cells as uniform fine-grid refinement
- Constant training rainfall produces a flat rainfall-conditioned response

Abstract

Extreme rainfall increasingly overwhelms urban drainage and produces pluvial flooding with limited warning, and city-scale screening depends on representations that are computationally scalable, updateable, and evaluated with explicit control for spatial dependence. Many data-driven susceptibility methods rely on proprietary damage or insurance labels, and random train-test splits can inflate skill when spatial dependence is ignored. To address these constraints, this study evaluates an open-label machine-learning framework that uses H3 as the common spatial support for learning, spatially blocked evaluation, and resolution control. Multi-source public flood observations are assembled on H3 cells, gradient-boosting classification and continuous-risk regression are fitted, and entire parent cells are withheld during H3-block spatial cross-validation. Under this blocked evaluation, a small Manhattan pilot (n = 141, 80% positive) achieves spatial cross-validation accuracy 0.784 ± 0.069 and F1 0.866, below the always-positive baseline (0.808 and 0.893); pooled out-of-fold ROC-AUC is 0.68 and average precision is 0.86. In the larger pilot (n = 956, 47.9% positive), accuracy is 0.642 ± 0.148, exceeding the constant majority-class baseline (0.521), with continuous-risk R^2 of 0.525 ± 0.112 and out-of-fold ROC-AUC of 0.70. Selective refinement uses about 57% as many cells as uniform R11 refinement, while the fitted model output is formalized as the rainfall-conditioned cell index PFI_h(c,r). Because rainfall is constant in the training data, the evaluated pilots support the learning and spatial-evaluation architecture; citywide skill and rainfall-conditioned discrimination remain unevaluated.

Keywords: pluvial flood; H3; discrete global grid; spatial cross-validation; machine learning; flood susceptibility.

1. Introduction

Urban flooding occurs predominantly during intense rainfall in densely built areas with limited drainage, and the pluvial form—flooding that develops when rainfall overwhelms drainage before entering watercourses—can appear rapidly and with little warning [1]. Assessing this hazard at city scale requires representations that are computationally scalable, updateable as new observations arrive, and evaluated in a way that does not inflate performance. Two limitations recur in data-driven pluvial-flood screening. The first is that some data-driven pluvial-flood indices rely on proprietary damage or insurance records, so the underlying labels are neither public nor transferable to jurisdictions without comparable records. The second is evaluation: random train-test splits can overestimate generalisation when spatially proximate observations occur in both the training and test sets because of spatial autocorrelation. For city-scale screening, these constraints create a spatial-representation problem as well as a modelling problem: observations, predictions, and evaluation need a spatial support that remains coherent as scale changes.

Discrete Global Grid Systems, and the hexagonal H3 system in particular [2], provide a scalable spatial substrate for integrating observations, predictions, and multi-resolution analysis. H3 is a hierarchical, predominantly hexagonal spatial index with neighbourhood operations and parent–child relationships across resolutions [3]. Hexagonal DGGS have been applied to multi-scale flood mapping under climate scenarios [4], and a closely related line of work aggregates a pre-existing machine-learning building-level pluvial susceptibility index into H3 cells to reduce query cost and expose resolution-dependent hotspot loss [5,6]. In that formulation, H3 serves primarily as a multi-resolution aggregation and communication layer for an inherited index, leaving open the question of how the same hierarchy might support model fitting and evaluation across unseen spatial blocks.

These limitations motivate an H3-native learning and evaluation architecture in which the grid acts as the common spatial support rather than only as an aggregation layer. Open flood observations are assembled directly on H3 cells, coarse H3 parents define the spatial holdouts, and the same hierarchy is subsequently used to examine scale effects and guide selective refinement. The contribution therefore lies in linking label construction, spatial validation, and resolution control within one hierarchical spatial reference; each of these components is established separately in prior work [4,7,8,9,10]. Throughout, PFI_h(c,r) denotes the fitted model's rainfall-conditioned flood-probability output for an H3 cell, distinct from both feature-importance measures and the H3-aggregated building index of Svellingen et al. [5].

Accordingly, the analysis asks whether predictive performance persists when entire H3 parent blocks are withheld relative to class-prevalence baselines; how hotspot membership changes as fine-scale labels are aggregated and whether trained cell scores can concentrate fine-resolution representation; and whether the fitted cell-level output can be formalised for rainfall conditioning when observed rainfall variation becomes available. These questions are examined on a smaller and an expanded Manhattan pilot extent rather than on a citywide scale.

2. Study area and data

The analysis uses two Manhattan pilot extents. The smaller is a Lower Manhattan bounding box (approximately 74.02–73.97°W, 40.70–40.76°N); the larger expanded-Manhattan box spans approximately 74.03–73.94°W, 40.68–40.80°N. Both lie within New York City and are described as pilot extents throughout.

The predictor and observation layers are drawn from public datasets downloaded in August 2026. Elevation is from the USGS 3D Elevation Program [11]; impervious fraction from the National Land Cover Database [12]; hydrography from NHDPlus High Resolution [13]; flood labels from the New York City Department of Environmental Protection stormwater flood polygons [14], NYC 311 flooding service requests [15], and USGS Hurricane Ida high-water marks [16]; and a negative control from FEMA Sandy storm-surge inundation [17]. Version and download dates are recorded in a repository download manifest. Building footprints are used to compute building density. A distance-to-water proxy is derived from NHDPlus hydrography in a tidal and shoreline setting. Table 1 lists the layers and their roles.

Table 1. Open data layers and their roles in the framework.

Layer	Source	Form	Role
Elevation	USGS 3DEP	Raster	Elevation, slope, flow-accumulation proxy
Impervious surface	NLCD fractional impervious	Raster	Impervious fraction, urban land-cover flag
Hydrography	USGS NHDPlus HR	Vector	Distance-to-water proxy
Building footprints	NYC MapHub	Vector	Building density
Flood observations	NYC DEP stormwater, NYC 311, USGS Ida HWM	Vector	Flood-risk target (continuous and binary)
Negative control	FEMA Sandy surge inundation	Vector	Coastal-overlap diagnostic; never a training label
Rainfall condition r	Synthetic constant grid (75 mm/h)	Raster	Condition r; constant in the present pilots

The cell-level flood-risk target combines the available open observations. Polygon sources contribute an intersection area fraction (0–1); point sources contribute a per-cell count. The risk score is the maximum of the area fraction and the point-presence indicator, clipped to [0,1]. The binary flood-class target is then defined deterministically as flood_class = 1[flood_risk ≥ 1e−9], so any cell with positive flood evidence is positive. These are observed flood labels, not verified inundation ground truth: NYC 311 records in particular reflect reporting and mapping processes, and their biases have been documented previously [7,8]. FloodNet is supported as an optional label source, but no usable FloodNet observations are included in the present analyses.

Rainfall enters the framework separately as the condition r (rainfall intensity). The pilot rainfall is a constant synthetic input representing an Ida-like scenario (75 mm/h), not radar or gauge data; this choice is deliberate and is reported as a limitation in Section 5.

3. Methods

The method uses the H3 hierarchy as a common spatial reference while assigning distinct roles to the resolutions used for model fitting, spatial blocking, scale diagnostics, and adaptive refinement (Fig. 1).

3.1 H3 representation

Supervised modelling uses H3 resolution 9 (R9) as the training and evaluation support. Resolution 10 (R10) provides the fine-label support for the scale-loss diagnostic, with hotspots subsequently rolled to R9 and R8; model fitting remains at R9 throughout. Adaptive refinement is a post-training step that replaces selected R9 cells with their resolution-11 (R11) descendants. All H3 indexing and spatial joins use longitude–latitude coordinates (EPSG:4326). Metric quantities are computed on this support: cell areas use H3 native cell-area calculations, distance-to-water uses a great-circle (haversine) distance, and the terrain derivatives are computed on the raster grid and averaged zonally over each cell. The workflow is summarised in Fig. 1.

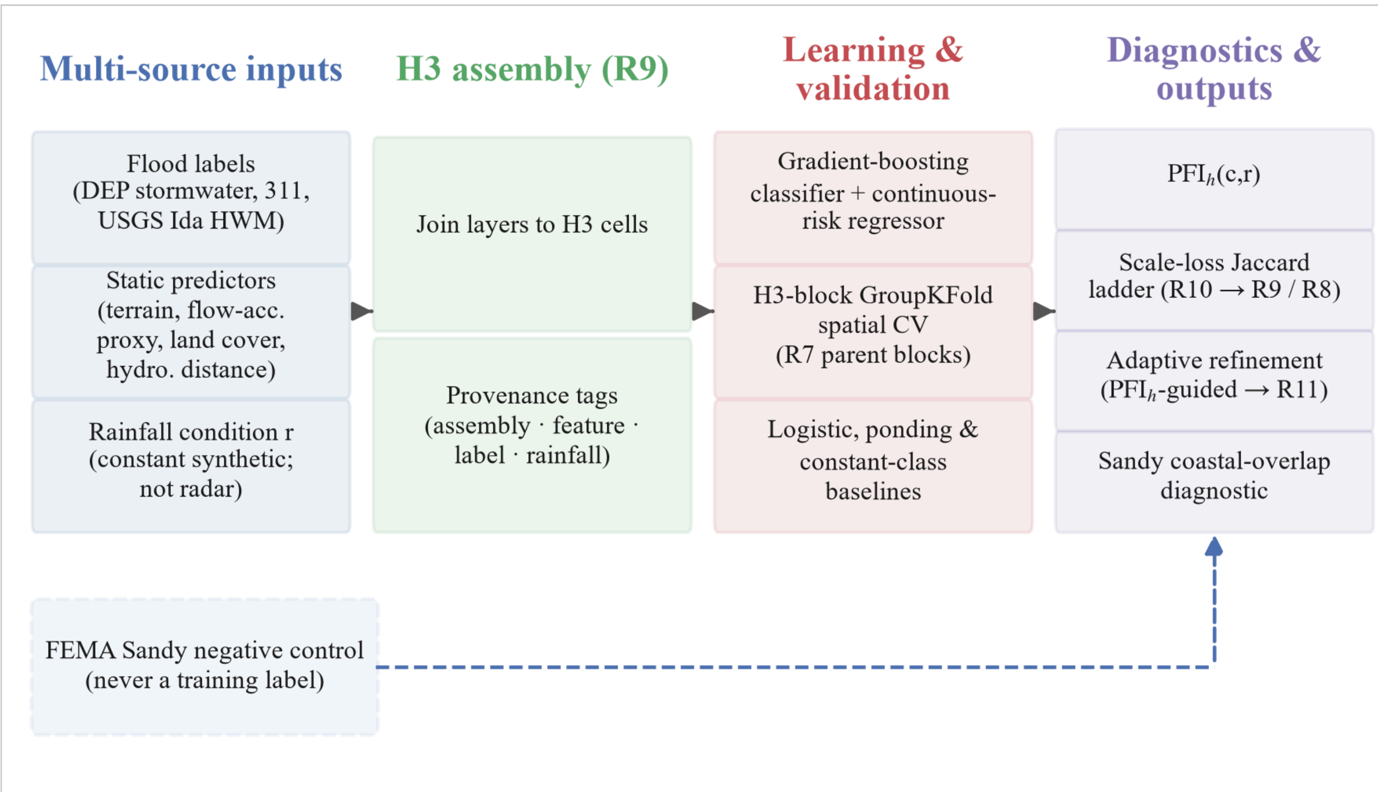

Figure 1. Open-label H3 pluvial-flood learning workflow. Open flood observations (DEP stormwater polygons, 311 flooding reports, and USGS Ida high-water marks), static predictors (elevation, slope, a flow-accumulation proxy, land cover, buildings, hydrologic proximity), and a rainfall condition r (a constant synthetic rainfall condition in the present pilot, not observed radar rainfall) are assembled into H3 cells at R9 with provenance tags. Gradient-boosting classification and continuous-risk regression are then evaluated under H3-block GroupKFold cross-validation against constant-class and logistic/ponding-rule baselines; diagnostics include the rainfall-conditioned PFI_h(c,r) index (currently flat scenario response), a scale-loss Jaccard ladder (R10 to R9/R8), adaptive refinement to R11, and a Sandy coastal-overlap diagnostic. The dashed FEMA Sandy side-channel represents the post-fit coastal-overlap diagnostic and has no role in model training.

3.2 Features

Static predictors are elevation, slope, a flow-accumulation proxy (D8-derived from the digital elevation model), impervious fraction (NLCD), an urban land-cover flag, building density, and distance-to-water (a shoreline/tidal-water proxy). Elevation, slope, and the flow-accumulation proxy are zonal means over each cell derived from the digital elevation model; impervious fraction is a zonal mean of the NLCD fractional-impervious surface; building density is the building-centroid count divided by cell area; distance-to-water is the great-circle distance from the cell centre to the nearest NHDPlus hydrographic feature; the urban land-cover flag marks cells whose impervious fraction exceeds 0.45. Rainfall is handled separately as the condition r rather than as a static feature.

3.3 Models and baselines

The primary learner is a gradient-boosting classifier and a continuous-risk gradient-boosting regressor, each with 80 estimators, maximum depth 4, and learning rate 0.08, fitted on features standardised within each training fold, with a fixed random seed (42); all other estimator parameters retain the scikit-learn 1.8 defaults. Two constant classifiers—always-positive and always-negative—are computed on each held-out fold, and accuracy and F1 are reported alongside these constant classifiers to provide a class-prevalence reference. The majority-class identity is determined from pooled target counts, and the constant-baseline accuracy and F1 use the same fold-wise aggregation as the model metrics. Two further baselines are included for diagnostic comparison: an L2-regularised logistic classifier and a ponding rule defined by the weighted combination

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0.40\cdot(1-\mathrm{elev}_{\mathrm{norm}}) + 0.35\cdot\mathrm{imperv} + 0.15\cdot(1-\min(\mathrm{slope},15^\circ)/15^\circ) + 0.10\cdot\mathrm{TWI}_{\mathrm{norm}},

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where elev_norm and TWI_norm are min–max normalised elevation and a topographic-wetness proxy, and a cell is classified positive when the score is at least 0.5. In-sample metrics are treated as optimistic references only. Table 2 summarises the models and baselines.

Table 2. Model and baseline specifications.

Model / baseline	Configuration	Role
Gradient-boosting classifier	80 estimators, max depth 4, learning rate 0.08, features standardised within each training fold, random seed 42	Primary binary learner
Gradient-boosting regressor	Same configuration	Continuous-risk learner
L2-regularised logistic classifier	scikit-learn 1.8 defaults	Diagnostic baseline
Ponding rule	Weighted combination of normalised elevation, impervious fraction, slope, and TWI; positive when score $\geq$ 0.5	Diagnostic baseline

Always-positive / always-negative	Constant class prediction	Prevalence-aware baselines on each held-out fold

3.4 Spatial H3-block cross-validation

Each R9 cell is assigned to its R7 parent, two H3 resolution levels coarser, and five-fold GroupKFold partitions these R7 groups so that no R7 block appears in both the training and the held-out portion of a fold. This changes the generalisation target from "new randomly drawn cells" to "new spatial blocks", which is closer to the transfer a screening product would face. For each held-out cell the predicted class probability is retained, so that threshold-independent discrimination metrics—ROC-AUC and average precision (AP), computed as the recall-weighted mean of precision across score thresholds [18]—are computed from pooled out-of-fold predictions. H3-block spatial cross-validation is the primary evaluation; random independent splits are retained as diagnostic comparisons.

3.5 Scale-loss diagnostics

Fine-resolution (R10) hotspot sets are defined by thresholding the open-label score at its 0.9 quantile and projected onto a coarser parent support; parent-aggregated (R9 and R8) hotspots are thresholded at the 0.9 quantile of the mean, maximum, and p90 rollups of the fine scores; the two sets are compared on that common parent support using Jaccard similarity and F1. The same diagnostics are additionally summarised through score distributions across resolutions and a pairwise hotspot-similarity matrix. Because many R10 open-label scores are tied at the maximum value, the 0.9-quantile threshold coincides with that maximum and the fine hotspot comprises every cell attaining it; the resulting counts are reported in Section 4.3. These diagnostics use different labels, resolutions, and hotspot definitions from the Jaccard value reported by Svellingen et al. [5] and are not a reproduction of that result.

3.6 Adaptive refinement

After training, the full-fit PFI_h probability screens R9 cells for refinement. A cell is selected when its PFI_h is at or above the 0.8 quantile of all cell scores, or when its predicted probability is uncertain (uncertainty $1 - 2|p - 0.5|$ of at least 0.7, i.e. p between 0.35 and 0.65); the selection is then expanded to include the one-ring H3 neighbourhood among the R9 cells. Each selected cell is replaced by its R11 descendants while unselected cells remain at R9. Model fitting precedes the refinement stage, which changes the spatial representation without altering the fitted R9 model. The primary metric is the resulting mixed-resolution cell count relative to a uniform R11 grid.

3.7 Rainfall-conditioned index

The cell index is defined as

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\mathrm{PFI}_h(c,r)=\widehat{P}(Y_c=1\mid X_c,r),

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where Y_c is the binary flood label, X_c are the static predictors, and r is the rainfall condition, and $\widehat{P}(\cdot)$ is the classifier's positive-class probability output. The classifier output is used directly without a separate calibration analysis. PFI_h is therefore the fitted cell-level model output and is distinct from both feature-importance measures and PFIb. Because the pilot uses a constant synthetic rainfall input, rainfall has zero training variance and the fitted PFI_h(c,r) is invariant across the evaluated rainfall scenarios; the definition is retained for subsequent analyses with observed rainfall variation.

3.8 Negative control

FEMA Sandy coastal inundation is excluded from feature construction, target construction, model fitting, and model selection. It is attached only after label assembly and used as a negative control that reports the overlap between pluvial evidence and coastal inundation (the coastal-only fraction) and the difference in mean observed flood-risk score between pluvial-only and coastal-only cells; Sandy labels are never used as training labels.

4. Results

4.1 Spatial pattern of labels, predictions, and the full-fit index

On the 141-cell Lower Manhattan R9 support, the observed open-label scores, out-of-fold probabilities, and full-fit PFI_h(c, r) show related but distinct spatial patterns (Fig. 2). The observed scores are bimodal by construction: cells with no positive flood evidence score 0, whereas positive evidence yields either a fractional polygon-overlap score or a point-presence score of 1 (median 1.0, mean 0.605; 84 of 141 cells at or above 0.8). The out-of-fold probabilities are on average high (mean 0.798) and less dispersed, consistent with the modest pooled discrimination reported in Section 4.2 (ROC-AUC 0.68 and average precision 0.86 at 80% positive prevalence). The full-fit index has a mean of 0.803 and shows moderate spatial concordance with the cross-validated surface (Pearson r = 0.51), as expected when the same cells, features, and labels are refitted without the five-fold holdout structure. The spatial concordance is descriptive; predictive performance is assessed separately from the out-of-fold metrics reported in Section 4.2.

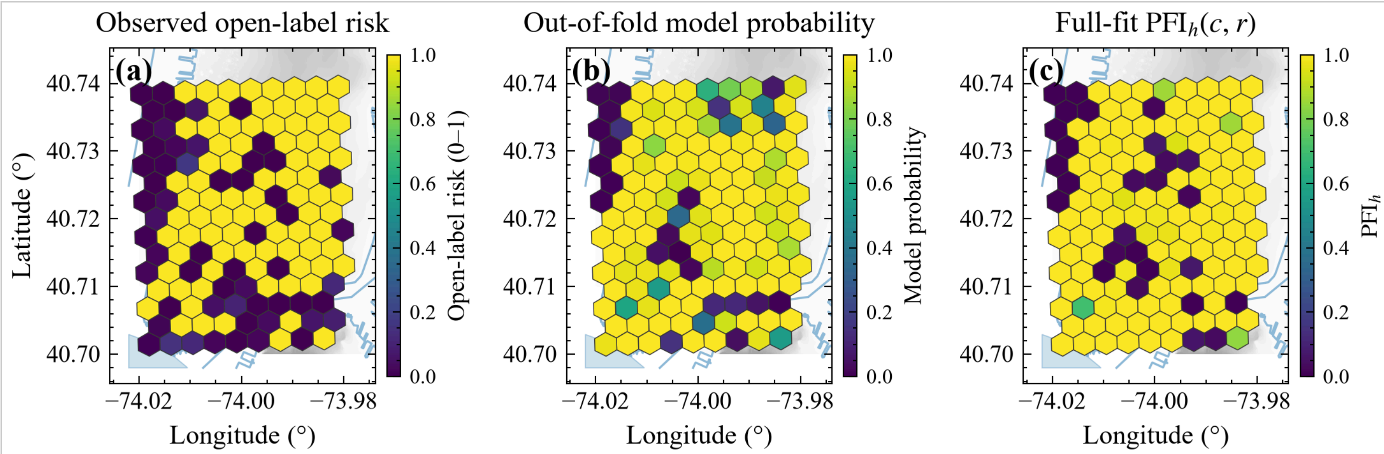

Figure 2. Spatial results for the Lower Manhattan pilot (n = 141 R9 cells). (a) Observed open-label flood-risk score; (b) pooled out-of-fold gradient-boosting probability under H3-block spatial cross-validation; (c) full-fit rainfall-conditioned index PFI_h(c, r) at the synthetic Ida-like condition r = 75 mm/h, fitted using all 141 cells. The surface is invariant across the evaluated scenarios because the training rainfall is constant (Section 4.5). Hexagons are coloured by value on a common 0–1 scale; grey shading is the DEM relief and light-blue lines/polygons are NHDPlus shoreline and water features.

4.2 Spatial H3-block cross-validation

The smaller pilot contains 141 R9 cells distributed over seven R7 blocks. Five-fold spatial cross-validation yields accuracy 0.784 ± 0.069 and F1 0.866 ± 0.044 (Fig. 3); per-fold accuracy/F1 are Fold0 0.755 / 0.850, Fold1 0.760 / 0.850, Fold2 0.773 / 0.872, Fold3 0.714 / 0.813, and Fold4 0.917 / 0.944. Here SD denotes the population standard deviation across the five held-out folds (ddof = 0). The held-out labels are highly imbalanced (80.1% positive), and a trivial always-positive classifier reaches mean accuracy 0.808 and mean F1 0.893 across the same folds—above the model's 0.784 and 0.866. The always-negative baseline accuracy is 0.192. Pooled out-of-fold ROC-AUC is 0.683, indicating modest ranking discrimination, and pooled average precision is 0.861, only slightly above the 0.801 positive-prevalence baseline. Continuous-risk R^2 is 0.030 ± 0.343 and is reported for completeness. Fold4 accuracy (0.917) coincides with a small test set (n = 24, two blocks) and is not interpreted in isolation. Table 3 summarises these numbers together with the expanded-pilot results.

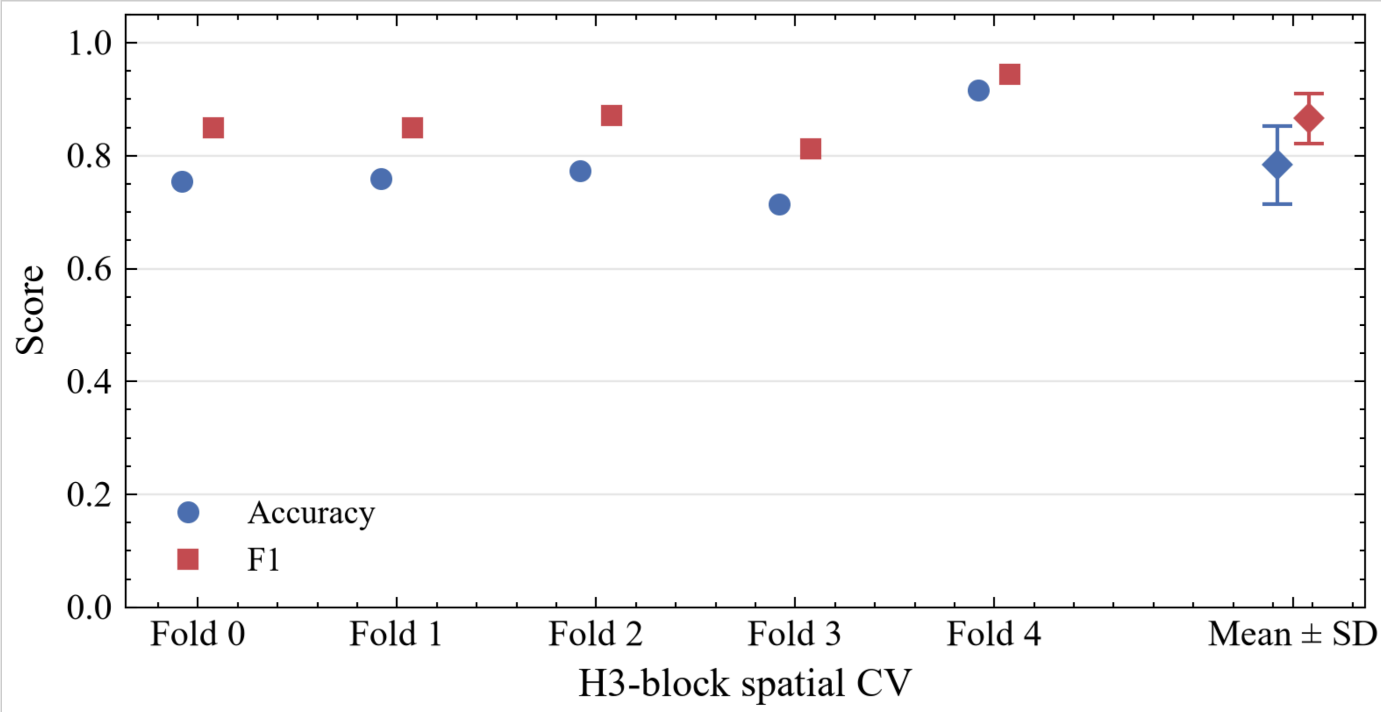

Figure 3. Spatial H3-block cross-validation performance for the Lower Manhattan pilot. Classification accuracy and F1 are shown as paired markers for each of five held-out folds formed from seven R7 H3 blocks (n = 141 cells); a final x-position shows the fold mean ± SD with error bars.

Table 3. Spatial H3-block cross-validation summary for the two pilots. Accuracy, F1, R^2, and MAE are fold means ± population SD across five held-out folds; ROC-AUC and average precision are pooled out-of-fold values. For the smaller pilot the positive class is the majority class (80.1%), so the always-positive classifier is also the majority baseline; for the expanded pilot the majority class is negative.

Metric	**Lower Manhattan (n = 141)**	**Expanded (n = 956)**
Accuracy	0.784 ± 0.069	0.642 ± 0.148
F1	0.866 ± 0.044	0.608 ± 0.185
Continuous-risk R^2	0.030 ± 0.343	0.525 ± 0.112
MAE	0.332 ± 0.074	0.112 ± 0.060
Pooled ROC-AUC	0.683	0.703
Pooled average precision	0.861	0.723
Always-positive accuracy	0.808	0.479
Always-positive F1	0.893	0.641
Always-negative accuracy	0.192	0.521
Majority-class F1	0.893 (positive)	0 (negative)

Note: SD denotes the population standard deviation across the five held-out folds (ddof = 0); the fold-mean values in the first four rows are arithmetic means of the per-fold metrics.

4.3 Scale-loss Jaccard ladder

Fine-resolution hotspots are defined at R10 by thresholding at the 0.9 quantile and rolled up to R9 and R8 under mean, maximum, and p90 aggregation (Table 4). The R10 hotspot comprises 571 of 991 fine cells because many fine open-label scores are tied at the maximum value, so the 0.9 quantile coincides with that maximum (Section 3.5). Under mean aggregation the R8 rollup yields Jaccard 0.167 and F1 0.286, whereas the R9 rollup yields 0.977 and 0.988. At R8, maximum and p90 aggregation both yield Jaccard/F1 of 1.000/1.000; at R9, maximum yields 1.000/1.000 whereas p90 yields 0.977/0.988. Maximum and p90 aggregation therefore retain more of the extreme signal than mean aggregation, but their higher overlap does not imply an absence of scale loss.

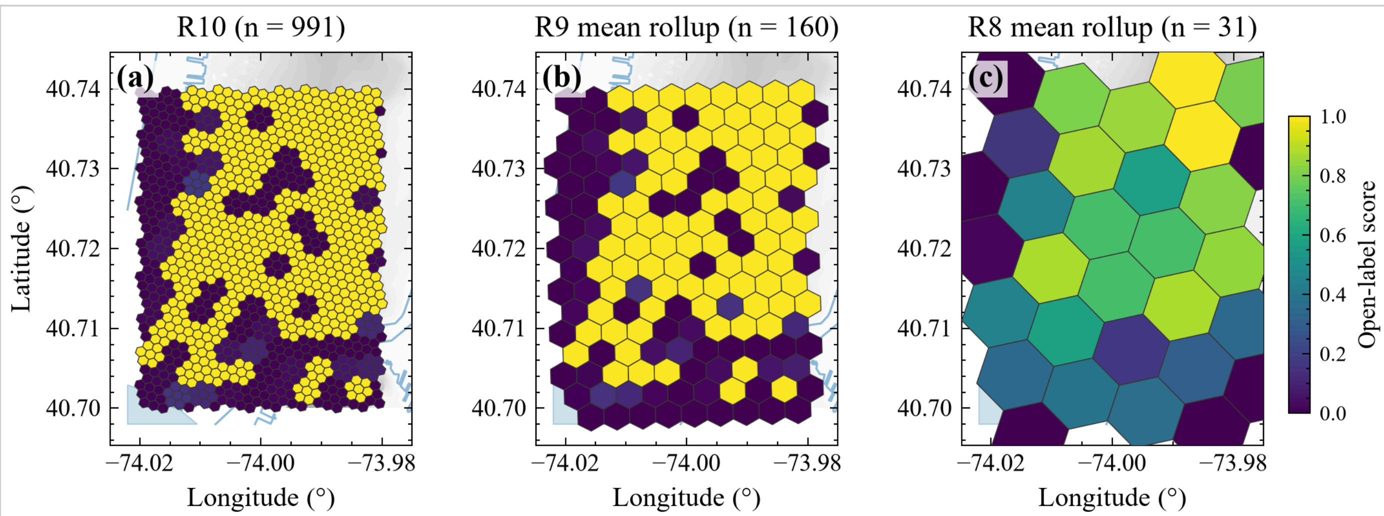

Figure 4. Multi-resolution open-label score surface on the label-assembly footprint. (a) R10 open-label flood-risk score (n = 991); (b) mean rollup to R9 (n = 160); and (c) mean rollup to R8 (n = 31). All panels use the same map extent and derive from the same R10 label-assembly footprint, with a common 0–1 colour scale that makes the smoothing accompanying coarsening visible directly. Figure 5 summarises cross-resolution hotspot similarity, while Table 4 quantifies sensitivity to the aggregation rule.

Fig. 4 maps the R10 open-label score surface and its mean rollups to R9 and R8 on the same label-assembly footprint, providing a spatial view of the smoothing that accompanies coarsening. Fig. 5a summarises the same scale dependence through the score distributions: the R10 distribution is wide and bimodal, while the mean rollups at R9 and R8 are progressively compressed, and Fig. 5b summarises the pairwise cross-resolution Jaccard similarity among the three resolutions, reproducing the R10-vs-R9 (0.977) and R10-vs-R8 (0.167) ladder values. Table 4 quantifies sensitivity to the aggregation operator. Figures 4 and 5 hold mean aggregation fixed: Fig. 4 shows the spatial effect of coarsening, whereas Fig. 5 summarises the corresponding changes in score distribution and hotspot membership. These values use different labels, resolutions, and hotspot definitions from the PFIb Jaccard of 0.14 reported by Svellingen et al. [5] and are not interpreted as a reproduction of that value. Table 4 lists the full ladder.

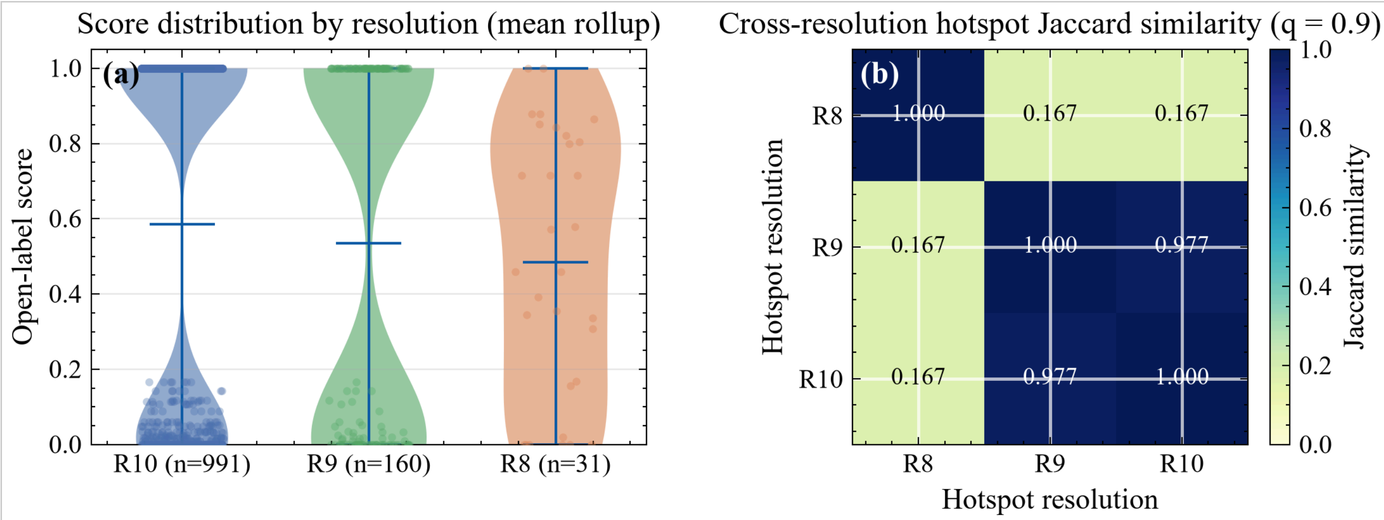

Figure 5. Resolution effects on the open-label score surface. (a) Violin plots with overlaid cell scores of the distribution at R10 (n = 991), R9 (n = 160, mean rollup), and R8 (n = 31, mean rollup), showing variance compression as the grid coarsens; internal bars mark the mean and extrema, and the distributions are descriptive summaries of the assembled scores. (b) Cross-resolution hotspot Jaccard similarity matrix (0.9-quantile thresholds) between hotspot sets at R10, R9, and R8, computed on the coarser support of each pair so the R10-vs-R9 and R10-vs-R8 entries reproduce the ladder in Table 4. The R10 label-assembly footprint contains 991 cells and aggregates to 160 R9 and 31 R8 parents, distinct from the 141-cell R9 supervised modelling table in Sections 4.1–4.2. For the realised hotspot sets, both comparisons involving R8 yield Jaccard similarity 0.167.

Table 4. Scale-loss ladder: hotspot Jaccard similarity and F1 between the R10 reference support and coarser representations under three aggregation rules (0.9-quantile thresholds; the fine R10 hotspot comprises 571 of 991 cells).

Coarse resolution	Aggregation	Jaccard	F1
R8	Mean	0.167	0.286
R8	Maximum	1.000	1.000
R8	P90	1.000	1.000
R9	Mean	0.977	0.988
R9	Maximum	1.000	1.000
R9	P90	0.977	0.988

Note: hotspot thresholds are empirical 0.9 quantiles of the assembled R10 scores; because many fine scores are tied at the maximum, the fine threshold coincides with that maximum and the R10 hotspot comprises every cell attaining it (571 of 991). The full ladder is plotted in Supplementary Fig. S1.

4.4 Adaptive versus fixed and uniform fine grids

The fixed coarse grid has 141 cells. Adaptive refinement selects 79 of 141 R9 cells and produces 3,933 mixed cells, compared with 6,909 cells for uniform R11 refinement (Fig. 6). The adaptive grid therefore uses 56.9% as many cells as the uniform fine grid and 27.9 times as many cells as the fixed R9 baseline. The ablation quantifies representation size by cell count; runtime, memory use, and city-scale computational cost were not evaluated. Table 5 lists the counts.

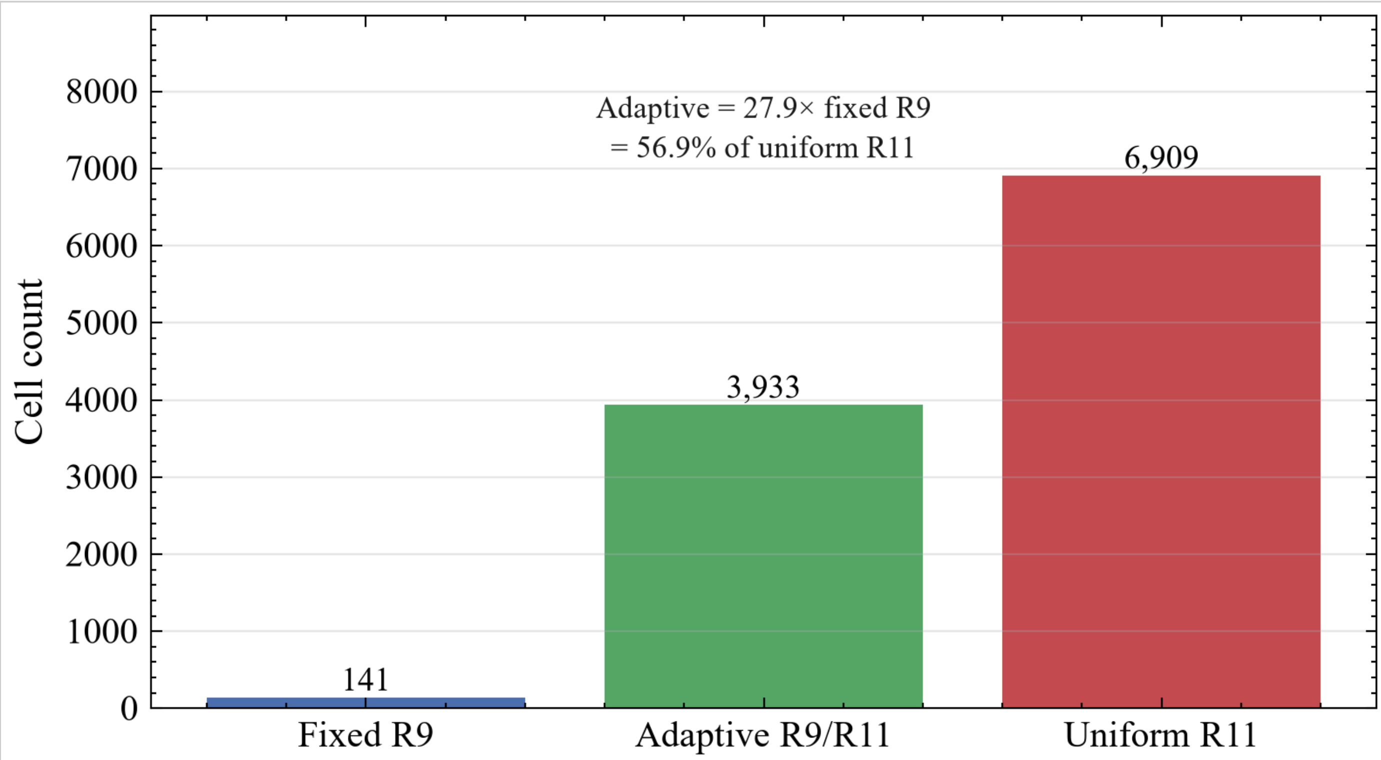

Figure 6. Adaptive refinement versus uniform fine grids by cell count. Fixed R9 (141), adaptive mixed R9/R11 (3,933), and uniform R11 (6,909) representations are compared for the Lower Manhattan pilot (adaptive = 27.9× fixed R9 = 56.9% of uniform R11; 79 of 141 R9 cells refined).

Table 5. Adaptive refinement versus fixed and uniform fine grids by cell count (Lower Manhattan pilot; adaptive = 27.9× fixed R9 = 56.9% of uniform R11; 79 of 141 R9 cells refined).

Representation	Cell count
Fixed R9	141
Adaptive R9/R11	3,933
Uniform R11	6,909

4.5 Rainfall scenarios

The scenario loop covers the 141 cells at four intensities—moderate (25), heavy (40), Ida-like (75), and extreme (100 mm/h). The mean PFI_h is about 0.803 for every scenario, and the within-cell range across scenarios is 0. Consequently, the fitted scenarios provide no rainfall-conditioned discrimination under the present training data. The flat response follows from constant training rainfall: every training cell carries the same synthetic value, so rainfall has zero training variance and contributes no learned variation to the fitted predictions. Evaluating rainfall responsiveness requires observed event rainfall with variation across intensities and model retraining.

4.6 Sandy negative control

Across the 141 cells, 31 intersect Sandy coastal inundation, 71 have pluvial evidence, and 23 have both; 8 cells are coastal-only (≈5.7%), and the mean observed flood-risk score is ≈0.120 higher in pluvial-only than in coastal-only cells. Coastal overlap is not a training label. Table 6 details the overlap and mean scores.

Table 6. Sandy negative-control statistics on the 141-cell pilot. The pluvial−coastal score difference is the mean observed flood-risk score of pluvial-only cells minus that of coastal-only cells.

Statistic	Value
Cells intersecting Sandy inundation	31 of 141
Cells with pluvial evidence	71 of 141
Both pluvial and coastal	23
Coastal-only	8 (5.7%)
Pluvial-only	48 (34.0%)
Neither	62 (44.0%)
Mean flood-risk score, coastal-only cells	0.500
Mean flood-risk score, pluvial-only cells	0.620
Mean flood-risk score, both	0.590
Mean flood-risk score, neither	0.613
Pluvial − coastal mean score difference	0.120

4.7 Expanded open-data pilot

The expanded open-data pilot contains 956 cells over 28 blocks, with near-even held-out labels (47.9% positive). Under the same H3-block protocol, spatial cross-validation accuracy is 0.642 ± 0.148, exceeding both the always-positive baseline (0.479) and the constant majority-class baseline (0.521, here the negative class); F1 is 0.608, below the fold-mean always-positive F1 of 0.641. Continuous-risk R^2 is 0.525 ± 0.112, MAE is 0.112, and pooled out-of-fold ROC-AUC is 0.703 with average precision 0.723, above the 0.479 prevalence baseline (Table 3). Per-fold accuracy/F1 are Fold0 0.801 / 0.832, Fold1 0.419 / 0.442, Fold2 0.759 / 0.736, Fold3 0.516 / 0.343, and Fold4 0.715 / 0.689. The per-fold spread coincides with heterogeneous held-out class composition—Fold1 and Fold3 are majority-negative, Fold0 and Fold4 majority-positive, and Fold2 nearly balanced (97/94)—and is not attributed to prevalence without further analysis. The expanded pilot provides a robustness check within Manhattan; citywide generalisation remains unevaluated.

5. Discussion

5.1 What the experiments establish

Taken together, the two pilots show two distinct effects of the H3 hierarchy: spatial blocking changes the interpretation of predictive performance, while coarsening and selective refinement expose a separate trade-off in spatial representation. The spatial patterns in Fig. 2 help explain this contrast: the observed labels are bimodal by construction, the model probabilities are high on average and less dispersed, and the full-fit index shows moderate spatial concordance with the cross-validated surface. Validation is based on the out-of-fold metrics. On classification the pilots differ: the smaller table does not beat its constant-class baselines on accuracy or F1, whereas the expanded table beats the constant majority-class baseline on accuracy (0.642 versus 0.521) but not on F1 against the always-positive comparator (0.608 versus 0.641). Pooled out-of-fold ROC-AUC indicates modest-to-moderate ranking discrimination in both pilots (0.68 and 0.70), and average precision is 0.86 and 0.72 respectively; because average precision is prevalence-dependent, the higher value in the smaller pilot is interpreted relative to its higher prevalence baseline. Accordingly, the results support measurable ranking discrimination but not strong thresholded classification, and no citywide classification skill is claimed.

5.2 Relation to prior work

Svellingen et al. [5] use H3 to aggregate a proprietary building-level index for scalable communication, whereas the present framework learns directly from public labels and evaluates predictions across held-out H3 spatial blocks. The R8 mean-aggregation Jaccard value of 0.167 is numerically close to their reported value of 0.14, but the two measures use different labels, resolutions, and aggregation procedures. The present ladder therefore characterises scale loss in the open-label analysis rather than reproducing their metric.

5.3 Methodological implications

Methodologically, the same H3 hierarchy links the spatial unit used for label assembly to the units used for cross-validation, scale analysis, and subsequent refinement, extending H3 from a post-prediction visualisation layer to the learning and evaluation architecture. The spatial cross-validation design is deliberately conservative, but one caveat applies: the R7 block size is fixed a priori; its relation to the target's spatial autocorrelation range was not evaluated, so block-size sensitivity remains a limitation.

5.4 Limitations

Several limitations constrain the interpretation of the results. The evidence is restricted to two sub-city Manhattan extents: Lower Manhattan (n = 141) and the expanded pilot (n = 956). The open labels, particularly 311 observations, reflect reporting and mapping processes rather than complete ground-truth inundation [7,8], while NHDPlus distance-to-water represents a tidal and shoreline proxy rather than inland drainage density. Independent FloodNet validation is also unavailable because no usable FloodNet observations are included.

Evaluation is further constrained by class composition and the number of spatial blocks. The smaller pilot is 80% positive and distributes seven H3 blocks across five folds (per-fold test size 21–49, with some folds containing a single block); the expanded pilot uses 28 blocks with per-fold test sizes of 190–193. The smaller pilot falls below the always-positive baseline, whereas the expanded pilot exceeds the majority-class baseline on accuracy but remains below the always-positive comparator on F1. Spatial cross-validation R^2 is 0.030 in the smaller pilot and 0.525 in the expanded pilot, and both values are interpreted only within their respective pilot extents.

Rainfall conditioning remains unevaluated under observed forcing. The current input is a constant synthetic rainfall value rather than event-specific gauge or radar rainfall, giving a within-cell PFI_h range of 0 across scenarios. Primary performance interpretation therefore rests on spatial cross-validation, with random-split accuracy retained as a diagnostic comparison. The fold-level variation, including Fold4 on n = 24 in the smaller pilot, further limits inference from any single fold.

5.5 Outstanding steps

Future work should first replace the synthetic rainfall condition with documented event observations and test whether PFI_h(c,r) varies across rainfall intensities for fixed static features. The same spatial cross-validation protocol can then be applied over broader, including citywide, extents and complemented by held-out FloodNet validation when a suitable sensor layer becomes available.

6. Conclusions

Using H3 as a common spatial support links open-label learning to spatially blocked validation, scale diagnostics, and selective refinement rather than treating the grid only as a post-processing layer. Across the two Manhattan pilots, the blocked evaluation and prevalence-aware baselines change the interpretation of classification performance, while selective refinement links resolution control to the fitted cell scores without requiring uniform fine-grid representation. The evidence remains limited to the evaluated pilot extents; observed event rainfall, rainfall-responsive predictions, citywide evaluation, and FloodNet validation remain priorities for further assessment.

CRediT authorship contribution statement

[待补充 — to be completed before submission: list each author with their CRediT roles.]

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Figure captions

Figure 1. Open-label H3 pluvial-flood learning workflow. Open flood observations (DEP stormwater polygons, 311 flooding reports, and USGS Ida high-water marks), static predictors (elevation, slope, a flow-accumulation proxy, land cover, buildings, hydrologic proximity), and a rainfall condition r (a constant synthetic rainfall condition in the present pilot, not observed radar rainfall) are assembled into H3 cells at R9 with provenance tags. Gradient-boosting classification and continuous-risk regression are then evaluated under H3-block GroupKFold cross-validation against constant-class and logistic/ponding-rule baselines; diagnostics include the rainfall-conditioned PFI_h(c,r) index (currently flat scenario response), a scale-loss Jaccard ladder (R10 to R9/R8), adaptive refinement to R11, and a Sandy coastal-overlap diagnostic. The dashed FEMA Sandy side-channel represents the post-fit coastal-overlap diagnostic and has no role in model training.

Figure 2. Spatial results for the Lower Manhattan pilot (n = 141 R9 cells). (a) Observed open-label flood-risk score; (b) pooled out-of-fold gradient-boosting probability under H3-block spatial cross-validation; (c) full-fit rainfall-conditioned index PFI_h(c, r) at the synthetic Ida-like condition r = 75 mm/h, fitted using all 141 cells. The surface is invariant across the evaluated scenarios because the training rainfall is constant (Section 4.5). Hexagons are coloured by value on a common 0–1 scale; grey shading is the DEM relief and light-blue lines/polygons are NHDPlus shoreline and water features.

Figure 3. Spatial H3-block cross-validation performance for the Lower Manhattan pilot. Classification accuracy and F1 are shown as paired markers for each of five held-out folds formed from seven R7 H3 blocks (n = 141 cells); a final x-position shows the fold mean ± SD with error bars.

Figure 4. Multi-resolution open-label score surface on the label-assembly footprint. (a) R10 open-label flood-risk score (n = 991); (b) mean rollup to R9 (n = 160); and (c) mean rollup to R8 (n = 31). All panels use the same map extent and derive from the same R10 label-assembly footprint, with a common 0–1 colour scale that makes the smoothing accompanying coarsening visible directly. Figure 5 summarises cross-resolution hotspot similarity, while Table 4 quantifies sensitivity to the aggregation rule.

Figure 5. Resolution effects on the open-label score surface. (a) Violin plots with overlaid cell scores of the distribution at R10 (n = 991), R9 (n = 160, mean rollup), and R8 (n = 31, mean rollup), showing variance compression as the grid coarsens; internal bars mark the mean and extrema, and the distributions are descriptive summaries of the assembled scores. (b) Cross-resolution hotspot Jaccard similarity matrix (0.9-quantile thresholds) between hotspot sets at R10, R9, and R8, computed on the coarser support of each pair so the R10-vs-R9 and R10-vs-R8 entries reproduce the ladder in Table 4. The R10 label-assembly footprint contains 991 cells and aggregates to 160 R9 and 31 R8 parents,

distinct from the 141-cell R9 supervised modelling table in Sections 4.1–4.2. For the realised hotspot sets, both comparisons involving R8 yield Jaccard similarity 0.167.

Figure 6. Adaptive refinement versus uniform fine grids by cell count. Fixed R9 (141), adaptive mixed R9/R11 (3,933), and uniform R11 (6,909) representations are compared for the Lower Manhattan pilot (adaptive = 27.9× fixed R9 = 56.9% of uniform R11; 79 of 141 R9 cells refined).

Supplementary Figure S1. Open-label hotspot scale-loss diagnostics across H3 resolutions. Jaccard similarity and F1 compare hotspot sets defined on the reference fine support, H3 R10 (0.9-quantile threshold), with R9 and R8 representations under mean, maximum, and p90 aggregation. This figure reproduces the numeric ladder reported in Table 4.

Data and code availability

The public repository (code, configs, tests, paper documentation, and small summary tables; large rasters, geojson, trained model binaries, and large parquet files are excluded) is available at https://github.com/Coucou2016/pluvial-flood-risk-DGGS-H3. The submission version is archived under the immutable tag `submission-v1`; the corresponding commit and provenance of all reported outputs are recorded in the accompanying audit document. Each raw layer is mapped in the repository download manifest to its source URL, retrieval date, and license. Analyses used scikit-learn 1.8.0 and H3 4.4.2 (exact versions are recorded in the run metadata). Synthetic demonstrations are excluded from scientific evidence. A process-oriented research report and a data-authenticity audit document accompany the manuscript in the same documentation folder.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT (OpenAI) as an editorial reviewer to review manuscript language, organization, and the presentation of scientific framing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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